Radar-Enhanced 3D Tree Trunk Reconstruction in Foliage-Occluded Forest Environments

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1. Motivation

The Viking Hills dataset is the first forest mapping dataset to feature co-registered LiDAR, camera, and radar data [1]. It was collected using a ground robot equipped with a 32-beam LiDAR sensor, a 4D imaging radar, and a camera. A key challenge is seasonal vegetation growth: in June, tall grass obstructs LiDAR measurements, causing trunk segmentation Intersection over Union (IoU) to drop by 7–14 percentage points compared to May [1]. In contrast, radar is less affected by foliage occlusion and can detect tree trunks beyond 30 m, although its limited lateral resolution (approximately 0.87 m at 40 m) makes precise trunk reconstruction difficult [1].

These complementary strengths and limitations motivate combining LiDAR, camera, and radar data to improve tree trunk reconstruction in densely vegetated environments. We investigate this approach through 3D Gaussian Splatting (3DGS) [2], addressing the following research question:

Does incorporating 4D imaging radar into a LiDAR-camera 3D Gaussian Splatting pipeline improve tree trunk reconstruction in foliage-occluded forest environments?

2. Related Work

Existing multimodal 3D Gaussian Splatting approaches primarily focus on autonomous driving and rely on two sensing modalities. Current 3D Gaussian Splatting (3DGS) frameworks, such as SplatAD [3], RaGS [4], and RadarSplat [6], have primarily been developed for autonomous driving applications. However, applying these approaches to forest reconstruction presents two key challenges:

- **Limited evaluation in forest environments:** These frameworks have primarily been developed and evaluated using autonomous driving datasets, focused on road-driving environments [3], [4], [6]. Forest environments introduce distinct challenges, including thin tree trunks partially obscured by dense grass and foliage, as well as sensor-specific limitations in detection range and visibility. Consequently, it remains unclear how effectively these approaches would generalize to forest environments, where different sensing modalities may observe different portions of the same scene.
- **Reliance on limited sensing modalities:** Existing approaches typically focus on one or two sensing modalities, such as cameras and LiDAR in SplatAD [3], or cameras and radar in RaGS [4]. While these combinations can provide useful geometric information for reconstruction in urban environments, their effectiveness in densely vegetated forests remains uncertain. Incorporating a third sensing modality could provide complementary geometric information, potentially improving reconstruction quality in regions where individual sensors are affected by occlusion or limited spatial resolution.

Existing 3DGS-based forest reconstruction methods primarily rely on cameras and LiDAR, with limited exploration of radar integration. Recent research on neural forest rendering has explored combining multiple sensing modalities. For example, Zhou et al. [5] proposed a 3D Gaussian Splatting framework that integrates handheld LiDAR point clouds with unmanned aerial vehicle (UAV) imagery to reconstruct forest environments. However, the integration of 4D imaging radar into 3DGS-based forest reconstruction remains relatively unexplored, particularly for reconstructing tree trunks under dense foliage occlusion.

3. Method

We propose building upon SplatAD [3], an open-source, real-time 3D Gaussian Splatting (3DGS) framework that integrates LiDAR point clouds and camera imagery, originally developed for autonomous driving scene reconstruction and rendering. We hypothesize that incorporating 4D imaging radar into SplatAD could improve tree trunk reconstruction in foliage-occluded forest environments. We propose two complementary approaches to integrating radar information into the existing framework:

3.1 Radar-Guided Gaussian Initialization

3DGS requires an initial set of Gaussians that are subsequently optimized during training [2]. Currently, SplatAD's Gaussian initialization methods rely on camera and LiDAR data [3]. However, its architecture allows for the addition of custom-initialized Gaussians. We propose developing a custom data parser that processes radar measurements from the Viking Hills dataset [1] to generate an additional set of Gaussians corresponding to tree trunks in regions where dense foliage attenuates or blocks LiDAR returns but remains detectable by radar. These radar-informed Gaussians will supplement SplatAD's existing initialization, potentially improving geometric coverage and reconstruction quality in heavily occluded regions.

3.2 Radar Loss Term: Vegetation-Aware Weighting

We define vegetation obstruction as the presence of vegetation along the line of sight between a sensor and a candidate 3D point. The Viking Hills dataset baseline [1] uses MinkowskiUNet [9] for radar point cloud semantic segmentation. For each radar point i, the standard cross-entropy loss is defined as

$$L_i = -\log p_i(y_i)$$

where $p_i(y_i)$ represents the model's predicted probability of the ground-truth class y_i. The overall baseline loss is computed as

$$L_{\text{base}} = \frac{1}{N}\sum_{i=1}^{N} L_i$$

where N is the total number of radar points.

We propose introducing a vegetation-aware weighting term to assign greater importance to trunk classification errors under dense vegetation conditions. Specifically, we define $w_i = 1 + \alpha V_i[y_i = trunk]$. where V_i is the vegetation obstruction score for point i, y_i equals 1 when the ground-truth class is trunk and 0 otherwise, and $\lambda \geq 0$ controls the weighting strength. The resulting weighted loss is

$$L_{\text{weighted}} = \frac{1}{N}\sum_{i=1}^{N} w_i L_i$$

This formulation increases the penalty for misclassifying tree trunks under dense vegetation conditions, encouraging the model to learn trunk features that remain discriminative despite foliage occlusion.

We consider two approaches for defining the vegetation obstruction score V_i.

Binary Vegetation Obstruction Score

The simplest approach is to define V_i as a binary variable based on the season in which the measurements were collected:

$$V_i = \begin{cases} 0, & \text{May (short grass)} \\ 1, & \text{June (tall grass)} \end{cases}$$

This formulation uses seasonal vegetation differences as a proxy for obstruction, assigning greater weight to trunk classification errors in June, when taller grass is more likely to obstruct sensor measurements [1].

Continuous Line-of-Sight Obstruction Score (Stretch Goal)

To obtain a more precise estimate of vegetation obstruction, we also propose a continuous line-of-sight obstruction score. Using the existing Viking Hills LiDAR point clouds and camera calibration, we construct a coarse 3D occupancy grid in which each cell c is assigned a vegetation classification:

$$G(c) = \begin{cases} 1, & \text{vegetation observed} \\ 0, & \text{non-vegetation or free space} \\ ?, & \text{unknown} \end{cases}$$

For each radar measurement i, we consider the ray r_i extending from the sensor to the corresponding 3D point. Let $N_v(r_i)$ denote the number of observed vegetation cells intersected by the ray, and let $N_k(r_i)$ denote the number of intersected cells with known classifications. We then define the continuous vegetation obstruction score as

$$V_i = \frac{N_v(r_i)}{N_k(r_i)}, \qquad N_k(r_i) > 0$$

This score ranges from 0 to 1, with higher values indicating a greater proportion of observed vegetation along the sensor's line of sight. Unlike the binary formulation, which assumes uniform vegetation obstruction within each season, the continuous score accounts for spatial variations in vegetation density and provides a more localized estimate of potential sensor occlusion.

For rays with no cells of known classification, the obstruction score is undefined and must be handled separately during implementation.

By incorporating this score into the weighted cross-entropy loss, we aim to improve radar-based trunk classification in heavily vegetated regions, potentially providing more reliable geometric information for subsequent 3D Gaussian Splatting reconstruction.

4. Evaluation

As a baseline, we first train our tree trunk reconstruction model using data collected in May from the Viking Hills dataset [1], when vegetation is relatively sparse and LiDAR measurements are less affected by foliage occlusion. We then render images from held-out camera poses excluded from training and compare them against the corresponding ground-truth images. Reconstruction quality is quantitatively evaluated using Peak Signal-to-Noise Ratio (PSNR) [7] and Structural Similarity Index Measure (SSIM) [8].

Next, we train the model on the same forest scene using data from both May and June to incorporate seasonal variations in vegetation density. We evaluate the model using held-out camera poses from June, when taller grass partially occludes tree trunks and reduces LiDAR visibility [1]. Our objective is to reconstruct the underlying tree trunk geometry despite these occlusions. We qualitatively assess the visibility and completeness of the reconstructed tree trunks and visualize the corresponding 3D occupancy grid to examine the reconstructed scene geometry and identify regions where occlusion may have affected reconstruction quality.

5. Project Timeline

We plan to complete the project over six weeks, with three team members working in parallel on data processing, model development, and evaluation.

Weeks 1–2: Dataset Preparation and Baseline Reconstruction. Set up SplatAD [3], preprocess the Viking Hills dataset [1], verify sensor calibration, and establish baseline reconstruction performance using May data.

Week 3: Radar-Guided Gaussian Initialization. Develop a custom radar data parser, integrate radar-informed Gaussians into SplatAD, and verify their contribution to trunk reconstruction.

Week 4: Vegetation-Aware Loss. Implement the proposed binary vegetation-weighted radar segmentation loss, with the continuous line-of-sight obstruction score as a stretch goal.

Week 5: Experimental Evaluation. Compare baseline and radar-enhanced reconstructions using May and June data, evaluating PSNR, SSIM, trunk visibility, and 3D occupancy grids.

Week 6: Final Analysis and Report. Conduct ablation experiments to assess the contributions of radar initialization and vegetation-aware weighting, analyze results, and prepare the final report and presentation.

References

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